

Soham Mondal

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Professional Summary

Computer Science & Engineering graduate (B.Tech, VIT) with published research in automotive predictive AI (Springer, forthcoming) and production experience across OEM technical solutions. Proven track record architecting full-stack client web portals, automating enterprise data pipelines with Python/Pandas, and integrating agentic workflows. Experienced delivering full-stack systems, telemetry analytics, and audit automation across global automotive accounts.

Technical Skills

Programming	Python, JavaScript, TypeScript, C++, Java, SQL, HTML5, CSS3
Web & Full Stack	React, Angular, Next.js, Node.js, REST APIs, Supabase, Tailwind CSS, RxJS, Bootstrap
AI & Agentic Tooling	Claude Code / CLI, Antigravity (AGY), OpenAI Codex, Higgsfield MCP, MCP Protocol/Servers, Agentic Workflows, scikit-learn, TensorFlow, XGBoost, SHAP, Pandas, NumPy, LSTM, Isolation Forest
Tools & Frameworks	macOS Terminal/CLI Integration, Git, Docker, Streamlit, Plotly, MySQL, MQTT, HiveMQ, Linux/Unix, KPI Dashboarding
Specialized	Vehicle Telematics, OBD-II Protocols, IoT Data Ingestion, Cross-Border Audit Automation
Design & UI/UX	Framer, Figma, Claude Design, Photoshop, Canva, Interactive Wireframing, Technical Documentation

Education

Vellore Institute of Technology (VIT)	Aug 2022 – Jul 2026
<i>B.Tech in Computer Science and Engineering</i>	<i>Convocated: Aug 2026</i>
Indian School Muladha, Oman	Jun 2020 – Jul 2022
<i>High School Diploma, PCMC</i>	

Professional Experience

MSX International (MSXI)	Gurgaon, India
<i>AI Solutions and Software Engineering Intern</i>	Dec 2025 – Present
◦ JSW Motors Portal: Architected an interactive single-page SOP web portal with modular component hierarchy, dynamic client-side Table of Contents (TOC) navigation, and fully responsive CSS architecture. ◦ TVS Norton Digital SOP Platform: Spearheaded end-to-end digitalization of the TVS Norton SOP workflow into an interactive full-stack web application, enabling dynamic procedure management and process automation. ◦ Volvo India & Global Auditing: Built automated Excel connection and ETL data pipelines to streamline cross-border dealership compliance auditing, eliminating manual spreadsheet consolidation. ◦ Client Deliverables (Porsche & VinFast): Updated and formatted SOP presentations and market-entry analysis decks for client reporting. ◦ Stellantis DPI & BMW IAM Analytics: Automated regional dealership KPI tracking and aftersales telemetry/inventory reporting using Python and Pandas, cutting monthly reporting turnaround from weeks to days. ◦ AI & Modern Tooling: Accelerated department engineering cycles by integrating Claude CLI, custom data parsers, and modern web scaffolding into standard development workflows.	
Mohsin Haider Darwish LLC (MHD ITICS)	Muscat, Oman
<i>Technical Intern</i>	May 2025 – Jul 2025
◦ Engineered enterprise data automation utilities in Python, optimizing warehouse reconciliation and workflow efficiency by 40% in an active production environment. ◦ Structured a centralized database schema managing 10,000+ inventory records and coordinated stakeholder reviews to enhance technical alignment.	
Febno Technologies	Remote, Dubai, UAE
<i>Angular Developer Intern</i>	Jul 2024 – Feb 2025
◦ Architected 25+ production Angular components using TypeScript, client routing, and dependency injection for web applications serving 1,000+ users. ◦ Integrated RESTful endpoints and orchestrated complex asynchronous state streams using RxJS, improving frontend response times by 30%.	

Research Publications

An Interpretable AI Approach for Predictive Vehicle Maintenance using XGBoost with Explainable AI ICADCML-2026 (Springer) *Forthcoming: Oct 20, 2026*

- Lead Author: Engineered an interpretable ML framework using XGBoost for component failure prediction (0.92 recall, 0.90 F1-score) and Remaining Useful Life (RUL) estimation (20.2h RMSE).
- Integrated SHAP-based Explainable AI to identify critical failure indicators; identified engine demand (28.4%) and coolant temperature (22.1%) as primary risk factors.
- Processed multi-source vehicle telemetry from 500 commercial vehicles, extracting 42 distinct health indicators from OBD-II sensor channels and CAN bus data.

Quantum Coin Flipping with Agent-Based Execution and Privacy Logging IEEE Access *Under Review*

- Developed a decentralized, provably fair coin-flipping protocol using Qiskit, agent-based execution, and quantum measurement properties.
- Implemented a privacy-focused logging framework using SHA-256 hash-chaining and cryptographic commitments for tamper-resistant auditability.
- Leveraged Hadamard-based superposition algorithms to achieve maximum entropy ($H = 1$ bit) for unbiased outcome generation in decentralized environments.

Projects

Predictive Vehicle Maintenance System using Machine Learning 2025

- Constructed a real-time vehicle maintenance prediction dashboard using Python, Streamlit, and TensorFlow with Isolation Forest anomaly detection and LSTM forecasting.
- Integrated OBD-II diagnostic signals (RPM, coolant temperature, engine load, speed, fuel pressure) with a two-layer LSTM (64/32 neurons) trained on 90 days of vehicle data for 7-day failure forecasting.

Cloud-Based Vehicle Tracking Simulation using IoT Protocols 2025

- Designed a scalable simulation for fleet tracking using MQTT and HiveMQ broker, addressing interoperability gaps in large-scale fleet telematics.
- Implemented a standardized JSON schema and topic hierarchy for structured telemetry ingestion across heterogeneous vehicle endpoints, visualized via a live Streamlit map interface.

Blockchain for Connected & Autonomous Vehicles 2025

- Pioneered a blockchain security architecture utilizing smart contracts for Vehicle-to-Everything (V2X) communication with tamper-evident event logging.
- Implemented on-chain and off-chain data partitioning strategies to minimize network latency while preserving cryptographic security and audit compliance.

Leadership Experience

SAE-VIT, Vellore Apr 2023 – May 2025

PR Head to Advisory Board

- Spearheaded marketing campaigns achieving 500,000+ social reach and managed technical automotive event logistics for 500+ participants.
- Mentored 20+ junior team members, expanding active club membership by 150%, and facilitated student technical workshops on vehicle electronics.

Additional Coordination: graVITas (Sep 2024) & Riviera (Feb 2024) event logistics for 5,000+ attendees; VIT Animation Club and Bulls & Bears Finance Club technical initiatives.

Certifications & Honors

Certifications: IBM AI & ML Specialization · Google Cloud Generative AI · Foundations of UX Design · Mobile App Development

Honors & Awards: Runner-up, Yantra'25 Binary Battles 2.0 (ISA, VIT) · 2nd Place, MOE-Oman Interschool Science Competition

Technical UI Systems: PrepVault App UI · Aushadhi Healthcare App UI · Minimalist Web Architecture

Languages

English (Professional) · Hindi (Native) · Bengali (Native) · Arabic (Elementary) · Japanese (Elementary)